

Assignment 6: 2D VOF Simulation of Bubble Rise

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October 2025

1 Governing Equations & Setup

The Volume of Fluid (VOF) method was used to simulate the rise of a single 6 mm air bubble in a quiescent water column. VOF solves a single set of momentum equations shared by the fluids, and the phase interfaces are tracked by solving the volume fraction continuity equation:

$$\frac{\partial \alpha_q}{\partial t} + \nabla \cdot (\alpha_q \vec{v}) = 0$$

Despite solving fewer equations than Eulerian-Eulerian approaches, VOF is computationally expensive because it requires extremely fine meshes and very small time steps to properly resolve and construct the sharp physical interface without numerical diffusion.

2 Courant Number Significance

The Courant number ($C = v\Delta t/\Delta x$) restricts the time step size relative to the mesh size. For explicit VOF formulations, C must generally be kept < 1 (often ~ 0.1) to ensure that the fluid interface does not cross more than one cell per time step, preventing severe numerical smearing and maintaining a sharp gas-liquid interface.

3 Bubble Deformation & Terminal Velocity

An explicit Geometric Reconstruction scheme with a Laminar viscous model was applied on a uniform grid (0.75 mm element size).

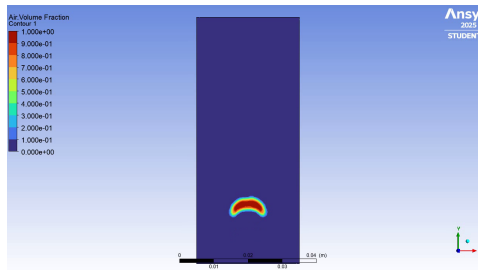


Figure 1: Deformed bubble tracking.

Initially spherical, the bubble rapidly deforms into a dimpled/skirted shape as it accelerates. Hydrodynamic forces (buoyancy vs drag) quickly overcome the surface tension forces holding it spherical. For a 6 mm bubble in water, the Morton number is $Mo \approx 5 \times 10^{-11}$ and the Eötvös number is $Eo \approx 4.89$. According to the Clift et al. (1978) bubble shape diagram, these non-dimensional parameters place the bubble exactly in the "wobbling/ellipsoidal" regime, matching the simulated deformation perfectly.

The vertical velocity rises sharply and eventually asymptotes as drag balances buoyancy, achieving the terminal rise velocity. Analytically, using wave-speed approximations for this regime, the theoretical terminal velocity is estimated at $U_T \approx 0.23$ m/s, which aligns closely with the asymptote observed in the simulation results.